

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Course Code: CBA1521

Course Name: Cloud Computing and
Big Data Analytics.

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CO2-AT2-Scenario-Based

Q1 Media Company - Cloud Video Streaming Platform.

Cloud Architecture:

- Use a Content Delivery Network (CDN) to cache videos near users.
- Deploy multiple application servers behind a load balancer.
- Store videos in cloud object storage.
- Use a distributed cache (Redis/Memcached) for frequently accessed data.
- Deploy services across multiple Availability Zones.

Load Balancing and Auto Scaling:

- Load balancer distributes traffic evenly among servers.
- Auto scaling automatically adds or removes servers based on CPU usage, network traffic, or user requests.
- This prevents server overload and minimizes buffering.

Cost optimization:

- Use auto scaling to avoid paying for idle resources.
- Store rarely accessed videos in low-cost storage.

- Use reserved instances for predictable workloads.
- Compress video files to reduce storage and bandwidth costs.

- Use CDN caching to reduce data transfer costs.
- Monitor cloud usage and eliminate unused resources.
- Purchase Reserve Instances.
- Select the appropriate VM size based on demand.
- Optimize data transfer between regions.

Q2) Fin Tech Startup - Multi-Cloud Containers:

Solutions:

- Use Docker containers for application packaging.
- Use Kubernetes for container orchestration across multiple cloud providers.
- Maintain the same deployment configuration in every cloud.

Service Discovery and Scaling:

- Kubernetes Service Discovery automatically locates services.

- Horizontal pod Autoscaler increases or decreases pods according to CPU or memory usage.
- Load Balancing distributes requests among available containers.

Benefits and risks of Multi cloud Deployment:

- High availability.
- Vendor lock-in is avoided.
- Enables workload distribution across providers.
- Protects against cloud provider outages.
- Allows selection of best services from each provider.
- Requires skilled administrators for management.
- Better disaster recovery.
- Improve scalability.

Risks:

- Complex management.
- High networking costs.
- Security policies must be synchronized.
- Increased monitoring complexity.

Q3) E-Learning platform - Backup and Disaster:

Solution:

- Enables automates cloud backups.
- Store multiple backups copies in different regions.
- Use object versioning to recover deleted files.
- Replicate databases continuously to a secondary region.

RTO and RPO:

- RTO (Recovery Time Objective): Reduce downtime by using automated failover and standby servers.
- RPO (Recovery point Objective): Reduce data loss using continuous replication and frequent backups.

Business Continuity Steps:

- Create regular backup schedules.
- Test disaster recovery plans periodically.
- Monitor backup health continuously.
- Maintain redundant infrastructure.
- Automate failover procedures.

Q4) Government Agency - Hybrid Cloud.

Hybrid Cloud Architecture:

- Store sensitive applications and data on-premises.
- Store non-sensitive applications in the public cloud.
- Connect both environments using VPN or dedicated private connections.

Identity Management and Encryption:

- Implement Identity and Access Management (IAM) with Role-Based Access Control (RBAC).
- Enables Multi-Factor Authentication (MFA).
- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.

Challenges:

- Complex infrastructure management.
- Compliance with government regulations.
- Network latency.
- Security policy synchronization.
- Higher operational costs.

Q5)

SaaS provider - High Availability.

Multi-Region Deployment Strategy:

- Deploy applications in multiple cloud regions. Use Global Load Balancer to route users to the nearest healthy region.
- Maintain identical application instances in every region.

Data Replication and Failover.

- Use synchronous or asynchronous databases replication.
- Configure automatic failover to the secondary region during failures.
- Perform regular failover testing.

Monitoring and Alerting:

- Monitor, CPU, memory, latency, application health, and network traffic.
- Configure alerts using CloudWatch, Azure Monitor, or Prometheus.
- Use centralized logging and dashboards.
- Enables automatic notifications through email or SMS.